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GenAI in the Context of African Universities: A Crisis of Tertiary Education or Its New Dawn?

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The rapid progression of generative artificial intelligence (GenAI) tools has raised significant interest and concern in academia. Instances of students submitting AI-generated assignments prompt investigations into implications for teaching, learning, and academic integrity. Recent publications highlight concerns such as a lack of conceptual understanding, threats to academic integrity, and disruptions to traditional assessment methods. While recognizing benefits like automated scoring and personalized learning, authors stress the responsible use of GenAI, emphasizing the educator's role in guiding students. This commentary identifies opportunities and threats of GenAI in African university contexts. Opportunities include increased operational efficiency, content generation, automated assessment, recognition of accessibility needs, overcoming language barriers, and accelerated research. However, these tools require human correction and cautious consideration of job displacement concerns. Threats encompass job displacement, privacy and security issues, threats to academic integrity, hallucinations/confabulations of GenAI, access and infrastructure challenges, technological overemphasis, lack of customization for local needs and cultural contexts, dependency on external providers, and unaffordable costs. The need for robust guidelines that balance technological advances with traditional teaching methods in African universities is emphasized. Given digital transformation initiatives like the African Union's Agenda 2063 and Botswana's SmartBots strategy, integrating GenAI could shape the future of African tertiary education. Proactive policies should address ethical concerns, ensure access, and make GenAI tools available, requiring a collaborative effort to navigate its impact responsibly.

CCS Concepts: • **Applied computing → Psychology;** • **Social and professional topics → Computing education;** • **Computing methodologies → Discourse, dialogue and pragmatics**;

Additional Key Words and Phrases: Generative artificial intelligence, universities, African countries

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1 Introduction

Generative artificial intelligence tools seem to be the next major thing after COVID-19 to completely change the world we live in. What was our first encounter with GenAI? We recall rather strange cases of Master's and PhD students submitting their draft proposals and dissertation chapters written in a smooth and grammatically correct manner, but generic, lacking specific detail, and, most importantly, not including references. When asked questions regarding the content of their manuscripts, students were not able to answer them. During the continuous assessment test (run on computers) some undergraduate students submitted surprisingly good answers to the essay-like question requiring a certain degree of creativity. Unfortunately, the answers submitted by several students were the same. It suddenly occurred to us that it might be one of those AI generative tools. We, of course, heard of, but never used these tools ourselves. We registered in ChatGPT and started exploring it immediately. In less than a minute ChatGPT generated, as per our queries, the following:

- terms of reference for the faculty research and innovation committee,
- the questionnaire to be delivered to the faculty members to collect their opinions on how research output can be improved,
- the memo requesting more funds to be allocated to the department, and
- the genuine recipe for a low-calorie dessert.

In all honesty, our first experience with GenAI was rather scary. It seemed like this new technology was going to overthrow almost anything we were led to believe previously (such as the importance of being able to write in a clear and literary manner or to produce a comprehensive literature review for dissertations and research articles). It was also unclear how we were to assess our students from now on. Is it their work or was it generated by GenAI? We felt a bit like math teachers protesting the use of calculators in classrooms in 1966 [1].

There were no university regulations regarding the use of GenAI tools by students to guide us. We assumed that most people in our university were not even aware of the existence of GenAI. Our subsequent steps involved, firstly, investigating the existence of any national or international policies or guidelines on the use of GenAI in education, and secondly, conducting a review of existing literature on the application of GenAI in university settings to ascertain the perspectives of other academics on this issue.

Given the significant push towards digital transformation in Africa, as evidenced by initiatives such as the African Union's Agenda 2063 [2] and the SmartBots digital transformation strategy in Botswana [3], the integration of GenAI in education could play a crucial role in shaping the future of African tertiary education. However, the lack of specific policies addressing GenAI use in African countries remains a gap. The potential for GenAI to address issues such as large class sizes, limited access to educational resources, understaffing, and overloading of teaching staff is immense. In the context of African universities, personalized learning solutions can address key educational challenges by providing tailored support to diverse student populations, optimizing resource use, and improving academic outcomes. By leveraging technology to create individualized learning experiences, African universities can enhance the quality of education and better prepare students for the demands of the modern workforce. Conversely, concerns about job displacement, privacy, and academic integrity are particularly pronounced in this context due to the existing socio-economic challenges.

2 Policies and Guidelines on the Use of GenAI Adopted in the African Context

At the global level, UNESCO recommends using ChatGPT in higher education with care and creativity, building the capacity to understand and manage AI, and conducting an AI audit to assess and monitor its ethical and effective implementation [4]. Currently, there is no policy regulating the use of GenAI (or even AI in general) in Botswana. In 2020, the Botswana Government adopted the SmartBots digital transformation strategy to create a smart sustainable society, focusing on public sector efficiency, education reform for a knowledge-based economy, and leveraging the 4th Industrial Revolution [3]. SmartBots was designed to accelerate digital transformation, including data-driven products and services that leverage digital technologies. Since the strategy was

implemented before the emergence of GenAI, SmartBots does not specifically address the use of GenAI, nor does it discuss AI in general.

Two African countries seemed to be the most promising in embracing the potential of GenAI, namely, Mauritius and South Africa. In the Government AI Readiness Index Ranking 2023, Mauritius leads the Sub-Saharan Africa region with a score of 53.27, followed by South Africa [5]. However, upon further investigation, we discovered that South Africa had been criticized for not putting enough effort into designing and adopting an AI strategy [6, 7], while three low-income (Rwanda) or lower-middle-income countries (Senegal and Benin) in 2023 adopted AI strategies. Before, the only country in the region with an AI strategy in place was Mauritius (an upper middle-income country) [8]. Provisions for AI were also made in the strategic plan entitled "Digital Mauritius 2030" [9]. Both the AI strategy and Digital Mauritius 2030 were adopted in 2018 and, for obvious reasons, did not mention GenAI. The analysis of more recent AI policies in three African countries demonstrated that none of them specifically dealt with the issues surrounding the use of GenAI [10–12].

A middle-income country like Botswana aiming to become a knowledge society should undoubtedly pursue the development of an AI policy or strategy, specifically addressing the use of GenAI in education. However, we assume that, to date, GenAI has not had any significant impact on Botswana society beyond academic circles. In developing the guidelines on ethical GenAI use, a bottom-up approach should be adopted. Instead of the government telling the universities how they should deal with GenAI, recommendations should come from academics who have firsthand experience with GenAI either through its usage or in dealing with students attempting to submit AI-generated texts as their original work. In developing national and regional policy guidelines for the use of GenAI in education, academicians will be able to distinguish between the impacts of GenAI on students, faculty, and educational outcomes, and to propose appropriate measures to address each type of impact.

3 Related Work

Although in the last few months, many articles relating to GenAI and its applications in various contexts have been written and are available online, those are mostly conference articlesand preprints indicating that GenAI as a research area has not been established yet. The earliest available manuscripts on the use of GenAI were published in January 2023. Below is a summary of what we have learned from these publications.

In [13], the authors point out that despite people widely using GenAI tools for text production and completely relying on them, these tools do not possess any conceptual knowledge or conscious understanding, as they only use word transitions and rely on probabilities of word classes. They might produce text that might look adequate at first glance but lacks truth and validity or refers to entities that do not exist. The authors inquire whether GenAI tools can be used for individual learning and the construction of knowledge. They are, however, not able to arrive at any decisive conclusion regarding the use of GenAI for teaching and learning, but rather hope that in the future their questions might be answered.

While [13] focuses on the potential of human-machine collaboration and whether learning with GenAI could be called collaborative, the article [14] concentrates on fostering trust and responsible use of GenAI in teaching and learning—the issues that are of the most interest to us at the moment. The author acknowledges the disruption of the entire landscape of education by GenAI. Same as in [13], there are more questions than answers. The difficulty of preserving academic integrity already has led to some universities completely banning the use of AI tools and others abandoning continuous assessment in favor of traditional paper-based, close-book final examinations. The author suggests that educators should fully embrace the potential of GenAI in teaching and learning while leading the students by example in the responsible use of these tools. The second recommendation is to teach students to understand the limitations of AI tools. Lastly, instructors should find areas where human intervention can add value to AI-generated output.

The review article [15] identifies applications of ChatGPT in education including automated scoring, teaching assistance, personalized learning, research assistance, quick access to information, generating case scenarios and exam questions, content creation for learning facilitation, and language translation. These listed applications

are very similar to those generated for us by ChatGPT (see Section 4). Supposedly the authors used GenAI to generate at least the data collection tool although they did not acknowledge it explicitly. The authors also discuss the challenges and limitations of using ChatGPT in education, such as its inability to reason beyond existing knowledge, generation of incorrect information, bias, potential undermining of students' critical thinking skills, and ethical concerns.

The article [16] focuses on the capability of academic staff to recognize the use of AI-generated context (by ChatGPT v. 4.0) in assessments. The results demonstrate that only approx. 55% of AI-generated submissions were recognized by faculty members while the AI detection tool was able to recognize approximately 91% of AI-generated content. The recommendation is to reconsider the weightings of assessments where GenAI tools can be used by students to imitate human-like responses. Those should be decreased (or even excluded completely) while assessments where GenAI tools cannot be applied should be allocated higher weights. Results presented in [17] demonstrate that the detection of the content generated by ChatGPT 3.5 is easier to detect than the content generated by ChatGPT 4.0. However, when applied to human-written control responses, the tools exhibited inconsistencies, producing false positives and uncertain classifications. This study underscores the need for further development and refinement of AI content detection tools as AI-generated content becomes more sophisticated and harder to distinguish from human-written text.

The impact of GenAI on US and Australian schools considered in [18] might not seem relevant to our commentary focusing mostly on university education. However, it provides evidence that the use of GenAI tools is not limited by institutions of higher learning, but starts much earlier, in grades 4–12 as also supported by [19] and [20]. Some schools in Australia banned the technology due to its potential "negative impact on student learning", while others—due to "the lack of reliable safeguards preventing these tools from exposing students to potentially explicit and harmful content" [21]. Globally, schools banned the technology amid fears that students could use it to generate convincing essay responses undetectable with current software [22].

Unlike other reviewed studies that have asked more questions than provided answers, [23] proposes recommendations on university guidelines regarding academic integrity and assessment design in the GenAI world. Their recommendations are based on the review of policies and guidelines of the leading world universities that implemented GenAI guidelines at the time of research (June 2023). The initial responses by many universities were to completely ban the use of GenAI, to control the environment in which students did their assignments, and to return to invigilated pen-and-paper tests. However, later, at least some of the universities developed GenAI policies and guidelines.

The three main themes identified by researchers [23] in university GenAI guidelines are *academic integrity, assessment design*, and *communication with students* on the use of GenAI.

The sub-themes within academic integrity are plagiarism, explicit acknowledgment of GenAI use, and detection of GenAI use. Using GenAI to generate entire essays, reports, or other written assignments or to rephrase plagiarized material is considered one of the forms of plagiarism. Some guidelines require explicit acknowledgment of GenAI use in student assignments but do not elaborate on how GenAI should be cited. Guidelines on the detection of GenAI mostly caution academics against fully relying on AI detection tools, as those are still in their infant stage and are unreliable.

The second theme provides recommendations to members of the academic staff regarding assessment design. First, they are advised to test their past and current assessments using GenAI tools to identify the capabilities and limitations of assessments. Having identified the weaknesses of the existing assessments, the instructors should re-design assignments in such a way that would prevent students from mindlessly copy-pasting the AI-generated content. The next recommendation is to focus on process and staged assessment design. An instructor, for example, might request students not only to submit an assignment but also to document the step-by-step process of preparing this assignment. Instructors are also advised to incorporate AI tools in the assessment design and use in-class assessments.

The third main theme, communication with students, suggests that students must be explicitly informed about the university's position regarding the use of GenAI. This can be done using module guides and open discussions in the seminars. The guidelines also should stipulate what exactly content (class policies, limitations of GenAI, the need for creativity, the importance of originality, etc.) is delivered to the students.

Authors of [23], however, assume that some of the reviewed guidelines might not have been applied yet in authentic university contexts and, therefore, should be tried and tested before putting them to use.

The only available publication discussing the use of GenAI in the African context reveals a significant gap in the awareness and acceptance of ChatGPT among Ghanaian academics [24]. The study found that most academics were initially unaware or had a limited understanding of ChatGPT and AI chatbots. Despite initial skepticism, the study reflected noticeable enthusiasm for the potential educational benefits of ChatGPT once academics were informed. These findings from Ghana mirror the situation in Botswana. In addition to opportunities, the article identifies threats associated with ChatGPT in education, including limited context understanding, inability to incorporate visuals, potential errors, facilitating academic dishonesty, cost-related inaccessibility, academic integrity issues, and concerns about bias and privacy. Out of these threats, however, only cost-related inaccessibility might be considered specific to the African context. This study underscores the importance of promoting technological awareness and adoption in African higher education to harness the benefits of innovative AI tools like ChatGPT.

4 Opportunities

We put a query to ChatGPT asking it about opportunities related to the use of GenAI tools in universities in African countries. The answer was remarkably good in its quality and similarity to a human expert's response. However, after carefully reading the AI-generated text, we found it rather generic and lacking specifics. Also, ChatGPT sometimes made obvious mistakes, such as not differentiating between GenAI and AI in general (such as proposing to use virtual or augmented reality to replace laboratory experiments). The revised list of opportunities and their potential positive impacts on university educational systems was linked to the context of the universities in African countries and supported by references to relevant publications. The opportunities suggested by GenAI are as follows:

Increased operational efficiency. Operational efficiency in the university environment can be enhanced through the automation of routine administrative tasks, focusing on producing drafts of various documents. Drafting, editing, and printing documents (memos, meeting minutes, agendas, etc.) take a significant amount of time and resources. Routine administrative tasks are typically performed by the administrative staff at the bottom of the organizational hierarchy (administrative assistants, personal secretaries, etc.). The percentage of administrative and support positions in African universities typically exceeds the percentage of academic staff (for example, 60% and 40%, respectively, but in some universities, the proportion could be 70% and 30%). The administrative personnel expenses are often too high, taking a large share of the limited university's resources. Another suggestion for increased operational efficiency is handling routine students' inquiries with AI chatbots, providing quick responses to students about schedules, course information, and other general queries. In universities in African countries, the number of students could be very large, but the number of administrative staff allocated to address their inquiries often is not enough.

Generation of learning content. Educational content (drafts of lesson plans, lecture notes, quizzes, assignments, etc.) can be created more effectively and efficiently using GenAI tools. However, it is necessary to emphasize that all AI-generated documents are only drafts requiring human correction and revision. The ability to quickly generate educational content is beneficial for university lecturers in developing countries, as they are often overloaded with teaching due to large classes and insufficient numbers of academic staff. A significant proportion of their workload is dedicated to the preparation for classes. As African universities strive to be innovative, research-based institutions of higher learning, lecturers are required to produce high-quality research

output in the form of publications in reputable journals. However, teaching leaves them very little time for research. Content creation with GenAI can help them reduce the time for preparation for classes and focus more on research, service to the university and community, and managerial duties. The use of GenAI helps to alleviate teacher workload and prevent burnout, with lesson planning and content development being significant contributors to teacher burnout [25].

Automation of assessment and providing feedback. We experimented with the assessment of essay-like test questions and assignments by putting the following queries to ChatGPT: "Compare the answer with the correct answer <correct answer> and grade it out of <maximum grade>: <given answer>" and "Check the answer to the question <question> and grade it out of <maximum grade>". The responses were of high quality and very much human-like. Not only were the marks provided, but also a comprehensive explanation of how the GenAI arrived at this grade. However, after some period of sheer excitement, we started noticing that ChatGPT was too "kind" and tried to find some meaning in the answers that were simply an attempt to guess. ChatGPT reduced the marks citing the lack of specific detail, allocating 1 out of 2 or even 1.5 out of 2 for the answers that deserved 0. However, because of the large number of students and overloading of most lecturers in African universities, this could be a good solution for marking open-ended test questions and essay-like assignments given that an assessor does not fully rely on GenAI.

Recognition of accessibility needs and inclusive content design. Providing speech-to-text or text-to-speech applications makes education more accessible for students with special needs. While most universities in African countries have adopted policies to support students with special needs, the practical measures are not yet implemented. Therefore, there is a gap between special needs policies in place and the practical implementation of them.

Overcoming language barriers in teaching and learning. Language translation tools and content generators allow access to educational content in a student's preferred language(s). In universities in African countries, the instructional language is often English, although for most students it is not their first or even second language. There are also international students from other African countries for whom language barriers might be a challenge. GenAI tools (e.g., chatbots) are recommended for learning new languages or improving the knowledge of a language of instruction by non-native speakers [26].

Acceleration of the research process. In the African context, leveraging ChatGPT can significantly accelerate the research process by providing data analysis capabilities, especially useful for universities with insufficient funds to purchase licenses for various software, and by aiding in the production of literature reviews, summarizing articles, and collecting information on research topics, thus enhancing understanding and efficiency in scientific writing while highlighting the importance of human oversight for accuracy and credibility. According to [27], the potential applications of ChatGPT in scientific writing include literature research, summarizing data, suggesting structure and titles, and aiding in the editing process. However, human guidance remains crucial for accuracy and credibility.

International collaborations. Similarly to overcoming language barriers in teaching and learning, generating draft documents (e.g., memorandums of understanding/agreement) in various languages and providing real-time translation services assists in fostering communication among researchers and other partners from different linguistic backgrounds. In many African universities, the official language of instruction and communication is English. However, for many of the staff, English is not a native language. Moreover, there are also African countries where the use of English is very limited or none (e.g., Angola or Mozambique). The study [28] highlights that the dominance of English as the primary language in science poses a significant obstacle to the full participation of non-native English speakers, particularly impacting early-career researchers.

5 Threats

Next, we put a query to ChatGPT asking it about threats related to the use of GenAI tools in universities in developing countries. Same as with opportunities, the response regarding threats was very general and lacked

specifics. The revised list of threats and their potential impacts was complemented by the authors' observations of the context of the universities in African countries and supported by references to relevant publications. The threats presented by GenAI are as follows:

Job displacement. The adoption of AI generative tools may lead to concerns about job displacement, especially for administrative roles that can be automated. This could affect the employment landscape in universities. Previously, we considered the automation of routine administrative tasks, such as preparing drafts of documents, as an opportunity to increase operational efficiency and decrease costs. However, while contributing to operational efficiency, the automation of administrative tasks could also cause job cuts, especially for those at the bottom of an organizational hierarchy. As the problem of unemployment is severe in many African countries, cutting jobs due to the replacement of people by GenAI tools can potentially lead to civil unrest. There is a strong cultural expectation in many African societies, including Botswana, to employ as many people as possible. As a result, even if certain jobs can be automated or replaced by technology, there is a cultural and social resistance to laying off employees. The example of low-paid, low-qualified jobs like security guards illustrates this well. Although CCTV cameras can replace the need for physical guards, these jobs still exist because they help support families and communities. Similarly, secretarial positions might be maintained to ensure people remain employed, even if GenAI can perform many of their tasks.

Privacy and security concerns. To better understand the privacy concerns presented by GenAI, we made an additional request to ChatGPT regarding the privacy threats presented by GenAI. The list of privacy concerns generated by ChatGPT includes deepfake technology, invasive data generation, manipulation of personal information, biometric data misuse, privacy violations in online platforms, data re-identification, lack of consent and awareness, surveillance and monitoring, and cross-domain inferences. The article [29] explores potential cyberattacks using GenAI tools, including social engineering, phishing, automated hacking, and malware creation. It also proposes defense techniques like cyber defense automation, threat intelligence, and ethical guidelines to enhance security measures. The article [30] emphasizes the need for legislation to address risks related to the provenance of AI-generated content, highlighting the importance of assessing human and machine-generated content differently in various contexts. In Africa, the privacy and security concerns associated with GenAI are especially severe due to limited cybersecurity infrastructure, insufficient regulatory frameworks, resource constraints, low digital literacy, inadequate awareness and training, rapid digitalization without safeguards, and dependence on external technologies.

Ethical concerns. The evidence suggests that more and more students turn to AI for learning, posing challenges to traditional teaching and assessment methods [14]. According to [31], approximately half of computer science students use GenAI tools for academic assignments. GenAI is widely used in undergraduate education, prompting the development of AI content detectors and classifier tools to differentiate between human-written and AI-generated content [17]. In [23] authors recognize the extremely fast adoption of GenAI by students due to its ability to generate human-like texts that are difficult to detect even by experts. Different sources report that 20 to 50% of students in different countries acknowledged the use of GenAI in their studies. The emergence of GenAI systems, particularly ChatGPT 4o, raises concerns about cheating and plagiarism due to its extensive knowledge base and adaptation to personal writing styles. A key concern of universities is that students may use GenAI to cheat or plagiarize their written assignments and exams [32]. This could undermine academic integrity and ultimately the reputation of universities. Given the novelty of this technology, the ethical concerns related to GenAI are relatively well-researched as compared to other threats. African students often face significant pressure to graduate quickly and enter the workforce. Unlike students in developed countries who might have the luxury of prolonged education and exploration well into their 30s, African students typically need to complete their education as swiftly as possible. This urgency stems from the necessity to start earning an income to support their families, including younger siblings and often single mothers who rely on them. African students are motivated to use any tools at their disposal, including GenAI, to ensure they graduate on time and with good grades, although not always in an ethical way.

Hallucinations. When asked to elaborate on what hallucinations in GenAI are, ChatGPT responds that those are instances where the AI model generates outputs that are inaccurate, illogical, or not grounded in reality, due to limitations in data processing and inherent biases in the training data. The article [33] focuses on the potential threat of producing "hallucinations" (inaccurate outputs), and the necessity to verify the accuracy of the generated content, specifically in research and education. The study [34] highlights the need for educators to recognize and decipher hallucinations and confabulations produced by GenAI. Hallucinations refer to instances where the GenAI generates false or nonsensical information, while confabulations are fabricated responses that seem coherent but are disconnected from reality. Students lacking critical thinking skills may easily mistake hallucinatory AI-generated content for authentic information, accepting it as truth. While this threat is not specifically unique to Africa, as students globally may lack critical thinking skills, the pressure to graduate with good marks and on time (e.g., having to submit several assignments to very tight deadlines) can exacerbate the reluctance of African students to verify AI-generated text before submission.

Access and infrastructure. In the African context, access and infrastructure present significant challenges in leveraging GenAI tools for educational purposes. Universities in African countries often struggle to provide consistent access to these advanced technologies due to inadequate funding and resources. This unequal access can exacerbate the existing digital divide between urban and rural areas, where infrastructure disparities are more pronounced. Additionally, the lack of reliable electricity and modern computing facilities in many regions further hinders the ability to integrate GenAI into higher education effectively.

Overemphasis on technology. There is a risk of overemphasizing technology at the expense of traditional teaching methods, potentially neglecting the importance of interpersonal skills and critical thinking, especially on the students' side. The study [35] highlights the necessity of combining technology with traditional teaching methods for a more effective learning experience. Authors suggest including evidence-based guidelines, fostering critical thinking and digital literacy skills, and promoting responsible GenAI use in higher education. As mentioned above, the pressure on African students to achieve high grades and graduate on schedule could lead to overemphasis on technology, causing them to overly rely on AI tools without developing the skills necessary to critically evaluate and cross-check information.

Lack of customization for local needs. Some GenAI tools may not be designed to address the specific needs and cultural contexts of developing countries, limiting their effectiveness in local educational settings. For example, in Botswana, where Setswana and other local languages are spoken, customization of GenAI tools for local languages is crucial for the adoption and use of e-government services (although it is less critical in the university environment).

Dependency on external providers. Universities relying on external vendors for AI solutions may face challenges related to vendor lock-in, rising costs, and potential service disruptions. For example, several months ago one of the authors attempted to upgrade her current free plan (GPT-3.5 only) to the Plus plan including more advanced tools such as GPT-4, DALL-E, and advanced data analysis. Unfortunately, she was put on the waitlist citing high demand and the inability of OpenAI to handle all upgrade requests. In the African context, this dependency is particularly problematic as universities already struggle with limited resources and funding. The reliance on external GenAI providers can intensify these challenges, making it difficult for institutions to access and maintain the necessary tools for educational and research purposes.

Unaffordable costs. Currently, students and most staff members in African universities utilize the free versions of GenAI tools like ChatGPT 3.5. One of the authors experienced a notable improvement in quality and functionality after upgrading to the Plus plan, which underscored the benefits of the paid version. In the future, all proprietary GenAI tools might require a subscription fee. The current monthly cost for ChatGPT 4o subscription is USD 20, a price that could be unaffordable for many students and staff in African countries. Additionally, universities without established policies or guidelines for GenAI may hesitate to cover these expenses.

Lack of reliable GenAI detection tools. This threat is closely linked to unaffordable costs. Reliable GenAI detection tools, if available, often come with a price tag that many universities are unwilling to cover due to

budget constraints. As a result, some lecturers might overlook students' unethical use of GenAI because they lack concrete evidence to prove assignments and research proposals were AI-generated. Accusing students without solid proof can lead to conflicts, with students potentially challenging these accusations and involving the university administration, given the high stakes for their academic success. This situation further complicates the integration and regulation of GenAI technologies in educational institutions in African countries.

6 Conclusion

The transformative potential of GenAI in reshaping higher education is significant, presenting both challenges and opportunities. The authors' initial encounters involved suspiciously generic submissions by students, leading to the realization of potential GenAI use. Uncertainty arises about how to assess students' work and the lack of university regulations on GenAI tool usage. This motivated the review of existing policies/guidelines and relevant literature and continuous experiments with GenAI tools (specifically, ChatGPT 4o and DALL-E). Existing studies highlight the limitations of GenAI, concerns about academic integrity, and the need for responsible use. GenAI applications in education include automated scoring, teaching assistance, content creation, and language translation.

In the context of African universities, the opportunities for GenAI include increased operational efficiency through the automation of administrative tasks, generation of learning content to assist instructors, automation of assessment and providing feedback to address workload challenges, recognition of accessibility needs and inclusive content design, overcoming language barriers in teaching and learning, acceleration of the research process, and facilitating international collaborations through language translation. Conversely, the threats encompass job displacement concerns due to the automation of administrative roles, privacy and security concerns, ethical concerns (e.g., regarding cheating and plagiarism), hallucinations and confabulations, challenges in providing access and infrastructure, overemphasis on technology at the expense of traditional teaching methods, lack of customization for local needs and cultural contexts, dependency on external providers, and unaffordable costs.

The integration of GenAI in African higher education requires a careful and balanced approach. While GenAI offers significant benefits, it is essential to establish comprehensive guidelines and ethical considerations to ensure its responsible use. Policies should be developed to address ethical concerns, ensure access and infrastructure, prevent overreliance on external providers, and make GenAI tools available to both staff and students. A collaborative effort is necessary to navigate the impact of GenAI responsibly and equitably, balancing technological advancements with traditional teaching methods to enhance the quality of education in African universities.

To address the challenges of GenAI use in African nations, it is crucial to propose comprehensive national and institutional agendas. These agendas should focus on developing clear policies for ethical use, investing in digital infrastructure, and implementing training programs to build digital literacy among educators and staff. Additionally, localized GenAI tools should be created to cater to the specific linguistic and cultural needs of African students and staff, supported by research and development to refine these applications. Public awareness campaigns are necessary to educate stakeholders about the benefits and risks of GenAI, alongside initiatives to ensure equitable access for all students. Finally, enforcing ethical guidelines will help prevent misuse and protect student data, ensuring responsible and fair use of GenAI in education.

References

[1] Rebecca Johinke, Robert Cummings, and Frances Di Lauro. 2023. Reclaiming the technology of higher education for teaching digital writing in a post-pandemic world. *Journal of University Teaching and Learning Practice* 20, 2 (2023).

[2] African Union. 2013. Agenda 2063: The Africa we want. *African Union, Addis Ababa, Ethiopia*, 2013. Retrieved August 30, 2024 from https://au.int/agenda2063

[3] Government of Botswana. 2020. SmartBots: Digital society reinvented. *SmartBots*, 2020. Retrieved August 03, 2024 from https://smartbots.gov.bw

[4] Emma Sabzalieva and Arianna Valentini. 2023. ChatGPT and artificial intelligence in higher education: Quick start guide. *UNESCO International Institute for Higher Education in Latin America and the Caribbean (IESALC)*. Retrieved August 30, 2024 from https://iesalc.unesco.org

[5] Emma Hankins, Pablo Fuentes Nettel, Livia Martinescu, Gonzalo Grau, and Sulamaan Rahim. 2024. Government AI Readiness Index 2023. Oxford Insights. Retrieved August 30, 2024 from https://oxfordinsights.com

[6] Simnikiwe Mzekandaba. 2024. South Africa must make moves on AI regulation. *ITWeb*. Published February 28, 2024. Retrieved August 29, 2024, from https://www.itweb.co.za/article/south-africa-must-make-moves-on-ai-regulation/P3gQ2qGANZk7nRD1

[7] Emile Ormond. 2023. Artificial intelligence in South Africa comes with special dilemmas - plus the usual risks. The Conversation. Published January 27, 2023. Retrieved August 29, 2024, from https://theconversation.com/artificial-intelligence-in-south-africa-comes-with-special-dilemmas-plus-the-usual-risks-197564

[8] Working Group on AI. 2018. The Mauritius artificial intelligence strategy. *Government of Mauritius, Port Louis*, 2018. Retrieved August 30, 2018 from https://ncb.govmu.org/ncb/strategicplans/MauritiusAIStrategy2018.pdf

[9] Ministry of Technology, Communication and Innovation of Mauritius. 2018. Digital Mauritius 2030. Government of Mauritius. Retrieved August 30, 2024 from https://ncb.govmu.org/ncb/strategicplans/DigitalMauritius2030.pdf

[10] Ministry of ICT and Innovation of Rwanda. 2022. The National AI Policy of Rwanda. Digital Watch. Retrieved from on August 30, 2024 from https://dig.watch/resource/the-national-ai-policy-of-rwanda

[11] D4DHub. 2023. Implementing a national AI policy: Key lessons from Rwanda. 28 November 2023. Retrieved March 13, 2024 from https://d4dhub.eu/fr/news/implementing-a-national-ai-policy-key-lessons-from-rwanda

[12] Ministry of Digital Affairs and Digitalization of Benin. 2023. National artificial intelligence and big data strategy. *Government of Benin, Porto-Novo*, 2023. Retrieved August 30, 2024 from https://dig.watch/resource/the-national-strategy-for-artificial-intelligence-and-big-data-sniam-of-benin

[13] Ulrike Cress and Joachim Kimmerle. 2023. Co-constructing knowledge with generative AI tools: Reflections from a CSCL perspective. *International Journal of Computer-Supported Collaborative Learning*, 18, 4 (2023), 607–614.

[14] Jonathan Sim. 2023. Embracing ChatGPT and other generative AI tools in higher education: The importance of fostering trust and responsible use in teaching and learning. *HESB* 2023, 13 (2023), 45–49.

[15] Bushra Mohammad, Tourjana Supti, Mahmood Alzubaidi, Hurmat Shah, Tanvir Alam, Zubeir Shah, and Mowafa Househ. 2023. The pros and cons of using ChatGPT in medical education: A scoping review. In *Studies in Health Technology and Informatics*. IOS Press, 2023, 644–647.

[16] Mike Perkins, Jasper Roe, James McGaughran, and Don Hickerson. 2023. Detection of GPT-4 generated text in higher education: Combining academic judgement and software to identify generative AI tool misuse. *Journal of Academic Ethics* 22, 1 (2023), 1–25.

[17] Ahmed Elkhatat, Khaled Elsaid, and Saeed Almeer. 2023. Evaluating the efficacy of AI content detection tools in differentiating between human and AI-generated text. *International Journal for Educational Integrity* 19, 1 (2023), 17.

[18] Lucinda McKnight and Troy Hicks. 2023. Generative AI writing tools: How they work, what they do, and why they matter. In *What PreK-12 Teachers Should Know About Educational Technology in 2023: A Research-to-Practice AnthologyPublisher: Association for the Advancement of Computing in Education (AACE)*.

[19] Tama Leaver and Suzanne Srdarov. 2023. ChatGPT isn't magic: The hype and hypocrisy of generative artificial intelligence (AI) rhetoric. *M/C Journal* 26, 5 (2023).

[20] Bethany Hiatt. 2023. National expert task force to be set up in bid to help Australian schools harness tools such as ChatGPT. The West Australian. Published February 28, 2023. Retrieved August 29, 2024, from https://thewest.com.au/news/education/national-expert-task-force-to-be-set-up-in-bid-to-help-australian-schools-harness-tools-such-as-chatgpt-c-9895269

[21] Louisa Clarence-Smith. 2023. Goodbye homework!' Elon Musk praises AI chatbot that writes student essays. The Telegraph. Published January 5, 2023. Retrieved August 28, 2024 from https://www.telegraph.co.uk/news/2023/01/05/homework-elon-musk-chatgpt-praises-ai-chatbot-writes-students/

[22] Caitlin Cassidy. 2023. Queensland public schools to join NSW in banning students from ChatGPT. *The Guardian* (2023).

[23] Benjamin Moorhouse, Marie Yeo, and Yuwei Wan. 2023. Generative AI tools and assessment: Guidelines of the world's top-ranking universities. *Computers and Education Open* 5, 2 (2023), 100151.

[24] Michael Adarkwah, Samuel Amponsah, Micheal Van Wyk, Ronghuai Huang, Ahmed Tlili, Boulus Shehata, Ahmed H. S. Metwally, and Huanhuan Wang. 2023. Awareness and acceptance of ChatGPT as a generative conversational AI for transforming education by Ghanaian academics: A two-phase study. *Journal of Applied Learning and Teaching* 6, 2 (2023), 1–16.

[25] Reem Hashem, Nagla Ali, Farah Zein, Patricia Fidalgo, and Othman Abu Khurma. 2024. AI to the rescue: Exploring the potential of ChatGPT as a teacher ally for workload relief and burnout prevention. *Research and Practice in Technology Enhanced Learning* 19, 1 (2024), 023.

[26] Thomas K. F. Chiu, Benjamin Moorhouse, Ching Sing Chai, and Murod Ismailov. 2023. Teacher support and student motivation to learn with artificial intelligence (AI) chatbot. *Interactive Learning Environments* (2023), 1–17.

[27] Michele Salvagno, ChatGPT, Fabio Taccone, and Alberto Giovanni Gerli. 2023. Can artificial intelligence help for scientific writing? *Critical Care* 27, 1 (2023), 75.

[28] Tatsuya Amano, Valeria Ramírez-Castañeda, Violeta Berdejo-Espinola, Israel Borokini, Shawan Chowdhury, Marina Golivets, Juan González-Trujillo, Flavia A. Montaño-Centellas, Kumar Paudel, Rachel White, and Diogo Verissimo. 2023. The manifold costs of being a non-native English speaker in science. *PLoS Biology* 21, 7 (2023), e3002184.

[29] Maanak Gupta, CharanKumar Akiri, Kshitiz Aryal, Eli Parker, and Lopamudra Praharaj. 2023. From ChatGPT to ThreatGPT: Impact of generative AI in cybersecurity and privacy. *IEEE Access* 99 (2023), 1–1.

[30] Alistair Knott, Dino Pedreschi, Raja Chatila, Tapabrata Chakraborti, Susan Leavy, Ricardo Baeza-Yates, David Eyers, Andrew Trotman, Paul Teal, Przemyslaw Biecek, Stuart Russell, and Yoshua Bengio. 2023. Generative AI models should include detection mechanisms as a condition for public release. *Ethics and Information Technology* 25, 4 (2023), 55.

[31] Ethan Dickey and Andres Bejarano. 2023. A model for integrating generative AI into course content development. 2023. Retrieved November 18, 2023 from https://www.researchgate.net/publication/373333352_A_Model_for_Integrating_Generative_AI_into_Course_Content_Development

[32] Cecilia K. Y. Chan. 2023. A comprehensive AI policy education framework for university teaching and learning. *International Journal of Educational Technology in Higher Education* 20, 1 (2023), 38.

[33] Cameron Atkinson. 2023. ChatGPT and computational-based research: benefits, drawbacks, and machine learning applications. *Discover Artificial Intelligence* 3, 1 (2023), 43.

[34] Tom Farrelly, Orna Farrell, Fiona Concannon, Eemon Costello, and Leigh Wolf. 2023. Editorial: There's an AI for that: Rhetoric, reality, and reflections on EdTech in the dawn of GenAI. *Irish Journal of Technology Enhanced Learning* 7, 1 (2023).

[35] Cecilia K. Y. Chan and Katherine K. W. Lee. 2023. The AI generation gap: Are Gen Z students more interested in adopting generative AI such as ChatGPT in teaching and learning than their Gen X and millennial generation teachers?. *Smart Learning Environments* 10, 1 (2023), 60.

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